

C.3 REQUIREMENTS

Supply items to be delivered, work to be performed and required deliverables shall be specifically described in individual delivery orders (DO), issued by the Contracting Officer and shall be within the parameters of one or more of the requirements described below. Hardware to be furnished hereunder may consist of any individual item or group of individual items related to the SPQ-15 capability. The Contractor shall perform and successfully complete all necessary testing including Factory Acceptance Testing (FAT) using the Government approved FAT procedures prior to delivery of all equipment.

The SOW describes the tasks required of the Contractor to support Naval Surface Warfare Center Dahlgren Division Dam Neck Activity (NSWCDD DNA) by providing services in the following areas:

- Production and Delivery (C.3.1)
- Engineering Services (C.3.2)
- Technical Services (C.3.3)
- Configuration Management and Integrated Logistics Support (C.3.4)
- Obsolescence Mitigation (C.3.5)
- Cybersecurity Support (C.3.6)

C.3.1 PRODUCTION AND DELIVERY

- a. The Contractor shall provide production and delivery of equipment and software to the Government based on approved Technical Data Packages (TDP) in accordance with industry standards.
- b. The Contractor shall correct all issues associated with Trouble Reports that are determined from testing or fleet related issues.
- c. The Contractor shall use an Engineering Change Proposal process for approval for any changes that are determined from testing for any produced radar distribution system according to the government approved TDPs.
- a. The Contractor shall conduct sufficient regression testing to ensure that any changes made appropriately correct the issues without introducing additional issues.
- d. The Contractor shall produce and deliver AN/SPQ-14(V) ASDS equipment, including any associated engineering changes, as listed in Section B.
- e. The Contractor shall produce and deliver AN/SPQ-15(V) DDS equipment, including any associated engineering changes, as listed in Section B.
- f. The Contractor shall configure all hardware and software components in strict accordance with the requirements documented in the applicable Defense Information Systems Agency (DISA) Security Technical Implementation Guides (STIG).
- g. The Contractor shall deliver all applicable DISA STIGs for all system hardware and software required to support and maintain the system's Authority to Operate (ATO).
- h. The Contractor shall formally document any STIG requirement that is not technically feasible to implement due to operational needs. This documentation must include a detailed justification for the deviation and shall be submitted to the Government for review and concurrence prior to implementation.

C.3.1.1 Warranties and Licenses

- a. The Contractor shall provide all production or repaired equipment, and material delivered under this contract with a warranty for 24 months after manufacturing completed date and Government receipt of equipment.
- b. The Contractor shall ensure that any equipment repaired or replaced under warranty successfully passes the same approved FAT stated under C.3.1.2 before it is returned to the Government. All testing under this clause shall be performed at no additional cost.
- c. The Contractor shall cover all labor and replacement parts for assemblies that have experienced failures that occur during the warranty period.
- d. The Contractor shall have the ability to repair any supported equipment at the Contractor's facility.
- e. The Contractor shall cover any shipping charges for any repaired equipment that failed during the warranty period.

C.3.1.2 Factory Acceptance Testing

- a. The Contractor shall develop and maintain the Factory Acceptance Test Plan and Procedures using the Original Equipment Manufacturer (OEM) Plan, Procedures and the approved TDP as guidance.
- b. The Contractor shall receive Government approval on any technical changes to the test plan or approaches to how the systems are being tested before making any changes to documentation or test systems. The Government will provide approval/disapproval within 10 business days.
- c. The Contractor shall conduct FAT on each version of the supported AN/SPQ-15(V) Unit and subsystems in accordance with Contractor- developed, Government-approved Acceptance Test Plan and Procedures.
- d. The Contractor shall document the results of FAT including the as-run, redlined, and marked-up procedures in a Qualification Test Report.
- e. The Contractor shall provide the Government updated procedures for any requested or required changes to the Acceptance Test Procedures.
- f. The Contractor shall provide all Test Data, evaluation tools (including simulations), developed hardware, developed software, prototypes, documentation, and results to the Government for independent evaluation of performance of any system produced or developed on this contract.
- g. The Contractor shall ensure all systems and software developed or delivered under this contract comply with all applicable and current DISA Security Technical Implementation Guides (STIGs).
- h. The Contractor shall perform FAT of software as required using Government approved test procedures.
- i. The Contractor shall perform FAT of hardware as required using Government approved Test Procedures.
- j. The Contractor shall be responsible for ensuring and demonstrating STIG compliance for all applicable operating systems, applications, network devices, and hardware that is associated with the supported systems.

- k. The Contractor shall notify the Government of planned test events no later than 15 business days prior to testing to support Government participation.
- l. The Contractor shall notify the Government of any changes to planned test events no later than seven (7) business days before the test.
- m. The Contractor shall reschedule the test at a mutually agreeable time within 15 business days in the event that the Government is not able to witness a test that requires Government attendance and/or approval.
- n. The Contractor shall schedule and conduct formal Systems Engineering milestone meetings at predefined project phases to review and discuss test results with designated Government representatives.
- o. The Contractor shall use a collaborative forum approach in these reviews to aid the Government to evaluate the progress, effectiveness, and compliance of the system under development, production, or Integration against established requirements and performance criteria.
The Contractor shall provide FAT Procedures to the Government for approval prior to conducting testing.

C.3.1.3 First Article Qualification Testing

- a. The Contractor shall conduct First Article Qualification Testing (FAQT) on the initial production units.
- b. The Contractor shall conduct FAQT on any production unit that has been modified from the original build-to-print configuration as a result of a configuration change.
- c. The Contractor shall submit the First Article Test report for all FAQT events.

C.3.1.4 Production Process

- a. The Contractor shall conduct In-Process Reviews (IPRs) to review the progress of the subsystem, software, system builds, integration, and testing efforts.
- b. The Contractor shall ensure the first In-Process Review (IPR) addresses the readiness of manufacturing process, quality systems, and production planning.
- c. The Contractor shall coordinate and deliver Meeting Agendas, Presentation Materials, and a record of Minutes/Actions Items in accordance with CDRLs XXXX and XXXX.
- d. The Contractor shall ensure the Production Status Reports include the production metrics as identified in the Manufacturing Plan (CDRLS XXXX and XXXX).
- e. The Contractor shall deliver Production Status Reports detailing metrics on production progress, labor and facility utilization, and material management, including inventory, expenditures, and quality control data. The Contractor shall provide a consistent format for the reporting of all metrics for each IPR.

C.3.1.5 Production Acceptance Test and System Burn-in Test

- a. The Contractor shall conduct a Production Acceptance Test (PAT) on each production AN/SPQ-15(V) equipment cabinet prior to Government acceptance (DD250).
- b. The Contractor shall include in the PAT the functional, electrical, mechanical and visual tests required to verify that the AN/SPQ-15(V) equipment is operational and works as specified in Government supplied requirements.

- c. The Contractor shall conduct the PAT at the Contractor's facility and utilize the Contractor test equipment in accordance with the Government approved PAT procedures.
- d. The Contractor shall provide Test/Inspection Reports for all acceptance tests.
- e. The Contractor shall conduct a System Burn-in Test (SBT) on each AN/SPQ-15(V) prior to Government acceptance.
- f. The Contractor shall operate the SBT for seventy-two (72) hours error-free. The Government reserves the right to witness the PAT.
- g. The Contractor shall provide a fourteen (14) business day notice prior to any SBT test event.
- h. The Contractor shall provide a test and inspection report in the event of a production test failure.
- i. The Contractor shall include in the Test/Inspection Report the circumstances of the failure, identification of the AN/SPQ-15(V) under test, the testing step conducted when the failure occurred, physical observations, and meter readings (as applicable).
- j. The Contractor shall provide an analysis of the test failure and corrective measures to prevent further test failures in the test and inspection report.

C.3.1.6 Test Readiness Review

- 1. The Contractor shall conduct a Test Readiness Review (TRR) prior to the execution of the FAQT. The TRR will address both the FAQT and FAT to ensure that the component, subsystem, or system under review is sufficiently mature to proceed into the major test cycle.
- 2. The Contractor shall conduct a TRR no later than 10 business days prior to the start of the FAQT.
- 3. The Contractor shall assess readiness for both the FAQT and the planned FAT to confirm the system is mature for the test cycle.
- 4. The Contractor shall review planned test objectives, methods, procedures, and statistical analyses, and shall examine lower-level test results, test plans, and test documentation to verify traceability of all planned tests to program requirements and to ensure that all required test resources are identified and coordinated at the TRR and Test Evaluation Review (TER).
 - a. The Contractor shall be responsible for the identification and coordination of all test resources required for project execution.
 - b. The Contractor shall propose all systems engineering reviews, as defined in the Program Executive Office, Integrated Warfare Systems (PEO IWS) Naval Sea Systems Command Research & System Engineering Warfare Systems (SEA 05) Technical Review Manual (TRM) for any production event.
 - c. The Contractor shall define the entry and exit criteria for TRRs as defined by the TRM.
 - d. The Contractor shall coordinate and send the Meeting Agenda to the Government prior to the TRR for review and approval.
 - e. The Contractor shall provide presentation materials, meeting minutes, and action items to the Government.

- f. The Contractor shall capture, document, and track Government and Contractor action items from the TRR and TER. The Contractor shall update Action Item status and closure plans monthly or as requested by the Government.

C.3.1.7 Packaging, Handling, Storage and Transportation

- a. The Contractor shall establish a Packaging, Handling, Storage and Transportation (PHS&T) program for the radar distribution system equipment, subsystems and parts developed under this contract.
- b. The Contractor shall plan, develop, and maintain the Packaging, Handling, Storage and Transportation (PHS&T) Plan to deliver the radar distribution systems equipment free-from-damage.
- c. The Contractor shall be consistent with the PHS&T program with the radar distribution system equipment principal component marking, reliability, maintainability, safety, and human factors.
- d. This Contractor will submit the PHS&T plan to the Government for approval before first product production for any product that is being produced for this contract
- e. The Contractor shall generate the radar distribution system principal component PHS&T procedures and elements that conform to MIL-STD-2073/1E and meet the requirements of Section D (Packaging and Marking).
- f. The Contractor shall be in accordance with best commercial practices.
- g. The Contractor shall document in the PHS&T plan any unique or special packaging, handling, storage or transportation requirements and state them in a PHS&T Impact Analysis Report.
- h. The Contractor shall document any applicable PHS&T requirements, PHS&T equipment designs, or procedures in the hard copy TM and Electronic Technical Manual (ETM)/Interactive ETM (IETM) source data, TDP, and provisioning data.
- i. The Contractor shall update the PHS&T Impact Analysis Report as required to support changes to the radar distribution system equipment implemented during production.

C.3.1.8 Item Unique Identification Marking

- a. The Contractor shall implement an Item Unique Identification (IUID) Marking Plan approved by the Government that meets Navy requirements for the AN/SPQ-15(V) Data Distribution System.
- b. The Contractor shall apply the IUID marking on each AN/SPQ-15(V) assembly in accordance with Defense Federal Acquisition Regulation Supplement (DFARS) 252.211-7003.
- c. The Contractor shall apply the IUID on Internal parts and subassemblies, unless designated as a Lowest Replaceable Unit (LRU), which are not required to have IUID marking.
- d. The Contractor shall submit all IUID pedigree data for LRUs embedded in unit level assemblies.
- e. The Contractor shall submit the IUID pedigree data into the IUID Registry as children to the parent unit.
- f. The contractor shall not submit the IUID pedigree data for any LRUs procured as spares shall not have a parent-child relationship.

- g. The Contractor shall Utilize MIL-STD-130 Series for IUID marking.
- h. The Contractor shall deliver preliminary IUID Marking Plan and IUID Marking and Verification Report.
- i. The contractor shall deliver final IUID Marking Plan.
- j. The contractor shall deliver final IUID Marking and Verification Report.
- k. The contractor shall provide a report that contains data resulting from IUID marking activities for all qualifying items manufactured on this contract.
- l. The contractor shall ensure the IUID marking data be available for inspection by the PCO or other Government representative at any time during the life of this contract.
- m. The contractor shall supply markings with data qualifier, data syntax, construct and semantics, as outlined in MIL-STD-130 Series, and utilize the IUID as the common data point for the marking of LRUs and enclosures.

C.3.1.9 As-Built Configuration Bill of Material List

- a. The Contractor shall create an As-Built Configuration (ABC) Bill of Material (BOM) List for each AN/SPQ-15(V) production unit, subsystem, and components, in accordance with the manufacturing plan annex, reflecting the hardware, software, and parts delivered with the AN/SPQ-15(V) production unit.
- b. The Contractor shall provide the BOM within the timeframe specified in the manufacturing schedule, with updates provided as necessary.
- c. The BOM shall serve as a key document for the warranty, maintenance, and lifecycle support of the radar distribution system production units. It will be used to facilitate repairs, parts replacement, and upgrades.

C.3.1.10 Delivery

- a. The Contractor shall deliver the radar distribution system Production Units in accordance with the PHS&T plan
- b. The Contractor shall use Special Packaging Instructions as outlined in the PHS&T plan.
- c. The Contractor shall update the Special Packaging Instructions as required in the PHS&T plan. The Contractor shall deliver all software (including source code), Software Development Life-Cycle (SDLC) files, and related hardware and firmware in a format compatible for installation onto Government-specified equipment.

C.3.2 ENGINEERING SERVICES

C.3.2.1 Systems Engineering

The Contractor shall develop and deliver documentation that demonstrates an understanding of detailed System of Systems requirements and Concept of Operations (ConOps) that the Government establishes to baseline a system.

- a. The Contractor shall use simulations, Analyses of Alternatives (AoAs), design trade studies, analytical efforts, study reports, models, and mock-ups to determine the feasibility of the proposed system; evaluate performance against requirements; assess cost and schedule viability; and identify potential technology needs and scope.

- b. The Contractor shall employ industry standards and methods to develop models for developed and supported systems including Model Based Systems Engineering (MBSE) and Model Based Product Support (MBPS).
- c. The Contractor shall perform design trade studies to evaluate alternative designs and identify the solution that provides the best combination of cost and effectiveness.
- d. The Contractor shall prepare a design trade study report that documents the evaluation methodology, assumptions, justification for the selected design, and consideration of Life Cycle Costing (LCC).
- e. The Contractor shall develop and deliver cost estimates in the form of Rough Order of Magnitude (ROM) estimates, vendor quotations, and detailed estimates informed by a Work Breakdown Structure (WBS) for the defined tasking.
- f. The Contractor shall develop a functional flow block diagram for the system and for all lower-level products that integrate into the system.
- g. The Contractor shall identify and document system-level qualities, properties, characteristics, functions, behavior, and performance.
- h. The Contractor shall identify and document all system interconnections for the top-level system, subsystems, and components.
- i. The Contractor shall design, modify, or reverse engineer systems to maintain interoperability and compatibility with existing systems.
- j. The Contractor shall complete detailed system and subsystem designs, fabricate hardware, develop software, and produce final designs for all system component end products.
- k. The Contractor shall develop and deliver a complete TDP that defines the system design configuration, performance requirements, and the procedures necessary to ensure item performance.
- l. The Contractor shall provide TDP review services on Government-specified TDPs to verify the accurate incorporation of all engineering, production, and logistics data. These services shall be performed when required to support Configuration Control Board (CCB) evaluations, in support of an approved Engineering Change Proposal (ECP) for a maximum of 10 instances per year. The Contractor shall develop and deliver technical documentation supporting the system's design and operation. This documentation shall fall into three primary categories: System Design Documentation: (e.g. system software/software architecture diagrams, interface control documents, specifications), User and Maintenance Documentation (e.g. System User Manuals, Maintenance Procedures, training guides), and Logistics and Packaging Data (e.g. packaging details, transportability reports).
- m. The Contractor shall provide all technical data associated with any hardware or software created or supported under this effort to the Government.
- n. The Contractor shall develop and maintain system models using Computer-Aided Design (CAD) tools, such as Autodesk Inventor or SolidWorks. This effort shall include generation of a complete design description for the selected solution; development of "make-to," "buy-to," and "reuse-to" specifications; and initiation of the development or acquisition of system products and enabling products.

- o. The Contractor shall establish and use formal interface management to maintain internal and external interface definition and compliance among the end-products and enabling products. This includes preparing interface management procedures, identifying interfaces, generating and maintaining interface documentation, managing changes to interfaces, disseminating interface information, and conducting interface control.
- p. The Contractor shall generate a specific product or systems through buying, making, or reusing to satisfy the design requirements. This includes building or coding the product; reviewing vendor technical information; and inspecting delivered, built, or reused products.
- q. The Contractor shall be responsible for the assembly and integration of validated lower-level end products into the higher-level end product. This responsibility includes the development of an integration strategy, detailed integration planning, acquisition and verification of integration-ready products, preparation of the integration environment, and development of product support documentation.
- r. The Contractor shall prove the end-product conforms to product requirements. This includes preparing for the verification efforts, analyzing the outcomes of verification (including identifying anomalies and establishing recommended corrective actions), and preparing a product verification report providing the evidence of product conformance with the applicable documents, to include Requirements Traceability Matrix (RTM), Systems Requirements Document (SRD), Software Requirements Specification (SRS), Hardware Requirements Specification (HRS), and Interface Requirements Specification (IRS).
- s. The Contractor shall ensure that the verified end product conforms to stakeholder expectations for its intended use within its intended environment and shall ensure resolution of all validation anomalies prior to transition. This effort includes preparation and execution of product validation, analysis of validation results (including anomaly identification and corrective action recommendations), and development of a product validation report providing objective evidence of conformance to the stakeholder expectations baseline.
- t. The Contractor shall host and conduct Production Acceptance Test and Evaluation (PAT&E) and First Article Testing of systems, equipment, components, and items to demonstrate compliance with all applicable requirements and specifications.
- u. The Contractor shall develop, implement, and maintain a technical risk management plan that provides for the continuous identification and assessment of risks associated with technical deviations from project plans. The plan shall include a technical risk management strategy; processes for risk identification, assessment, monitoring, and mitigation; and procedures for implementing mitigation and contingency actions when predefined risk thresholds are reached, in order to protect achievement of technical objectives across the project lifecycle.
- v. The Contractor shall continuously monitor and assess the progress of the technical effort and provide technical status information in support of system design, product realization, and technical management functions. This effort includes the development of technical assessment strategies and policies; assessment of technical work productivity and

product quality; tracking and trending of technical performance metrics; and execution of technical, peer, and lifecycle reviews.

- w. The Contractor shall participate in working groups as assigned from funded task orders.
- x. The Contractor shall identify the need for an ECP for existing systems and propose a recommended design solution to the Government. Following Government acceptance of the proposed ECP, the Contractor shall prepare, coordinate, and submit a final ECP package for Government approval.
- y. The Contractor shall ensure that all developed hardware and software incorporate a human factors development approach to support effective, safe, and usable operation.
- z. The Contractor shall deliver a detailed report that documents the application of the human factors' development approach and its incorporation into the changes delivered as part of the final solution.
- aa. The Contractor shall produce and deliver updates for existing variants of the AN/SPQ-15(V) system as directed by the Government.
- bb. Contractor implementation of the Functional Baseline (FBL) and Allocated Baseline (ABL) requirements involves the creation and release of engineering documentation that incrementally defines the configuration of the specific product.
- cc. The Contractor shall release engineering design data (engineering drawings, computer models, software design documents) to the Government.

C.3.2.2 Hardware Engineering

- a. The Contractor shall prepare and deliver ECPs, Notices of Revisions (NOR), Change Orders, Requests for Deviations (RFD), and/or Requests for Waivers (RFW) for Government review and approval.
- b. The Contractor shall implement ECPs once approved by the Government for any new development or legacy support for radar distribution products.
- c. The Contractor shall maintain Change Proposal information in a Government approved repository that is assessable by the Government.
- d. The Contractor shall update system/unit drawings including Installation Control Drawings as required.
- e. The Contractor shall provide Installation Test Procedures to the Government before testing.
- f. The contractor shall design and develop radar distribution components in accordance with specifications and engineering standards for installation. This shall include designs intended for use in either a lab environment or shipboard.
- g. The contractor shall design, develop, test, and support digital signal processing and radar distribution applications.
- h. The contractor shall support firmware and hardware development languages for embedded system design for microcontrollers, Field Programmable Gate Arrays (FPGA's), System on a Chip (SoC), and single board computers (VME, VPX, PXI, etc.).
- i. The Contractor shall develop and deliver TDPs defining all mechanical enclosures, masts, mounts, chassis, and structural supports required to install Commercial Off The Shelf (COTS) and Government Off The Shelf (GOTS) equipment in shipboard electronics racks. The contractor shall design and document cable assemblies, to

include cable lists; fabrication details; material specifications; wire pinouts; length; unique wire number identification; termination points; signal type(s); assembly weight; low smoke zero halogen (LSZH) status; cable label details; and build, inspection, packing and shipping instructions.

- j. The contractor shall design and develop power systems and TDPs for shipboard power distribution applications to support radar distribution applications.
- k. The contractor shall design and develop a power-based hardware solution that is used to solve and optimize different voltage and current requirements for component or system blocks for radar distribution applications.
- l. The contractor shall design and develop a TDP for rackmount electronics equipment. The TDP shall include assembly details, overall dimensions and weight, mechanical and electrical interfaces, material types, list of components, assembly instructions, fabrication details, label plate details, power requirements, cooling requirements, and electrical schematics.
- m. The Contractor shall design and develop all Printed Circuit Boards (PCBs) and Circuit Card Assemblies (CCAs) to meet Institute for Printed Circuits (IPC) Class 2 and Class 3 acceptance criteria.
- n. The Contractor shall develop solutions for obsolete or end-of-life (EOL) components or systems.
- o. The Contractor shall evaluate replacement components or systems to ensure equivalent or improved functionality and interoperability with interfacing components.
- p. The Contractor shall provide reverse engineering support for Government unlimited data rights by developing technical artifacts from existing designs to enable duplication through a complete set of technical drawings and documentation and shall verify and validate that the resulting replacement component functions as intended within the original system.

C.3.2.3 Software Engineering

- a. The Contractor shall develop new software versions to support emergent NSWCCD DNA requirements as directed by the Government.
- b. The Contractor shall develop updates for existing and new Software as directed by the Government.
- c. The Contractor shall provide Installation Test Procedures that adhere to DOD-STD-2106A to the Government for approval prior to conducting testing.
- d. The Contractor shall design network architectures, including networks that use virtualization suites, for systems. Network design shall include research, procurement, implementation, and testing.
- e. The Contractor shall design software architectures that define top-level requirements that enable allocated and derived requirements for lower layers of the physical and virtual software sub-systems and components to be developed and tested.
- f. The Contractor shall provide software engineering support to include the development, maintenance, testing, and Assessment & Authorization of software and operating system modifications for operating system integration efforts. Operating system integration activities will include the maintenance/modification of existing software code and

operating system configurations, establishment and design of new code, and implementation of enhancements to system capabilities.

- g. The Contractor shall follow the Application Security and Development Security Technical Implementation Guide (STIG) to maintain compliance for all developed software and provide completed checklist to the Government.
- h. The Contractor shall provide software engineering support to include development, maintenance, testing, and certification & accreditation of software for Human Machine Interface (HMI) Graphical User Interfaces (GUIs). GUI development activities include maintenance of existing source code, establishment and design of new code, and implementation of enhancement to system capabilities.
- i. The Contractor shall design & develop new, or evaluate existing, processing software for integration into systems. Digital processing will include converting one data format to another to support current and emergent standards for digital radar video formats.
- j. The Contractor shall support the C, C++, Java, and Python programming languages. Support for additional programming languages shall be provided as directed by the Government.
- k. The Contractor shall provide software engineering support to include the development, maintenance, testing, and Assessment & Authorization of software modifications for radar distribution applications and display.
- l. The Contractor shall develop software to facilitate operation of radar video distribution components in accordance with specifications and engineering standards for installation in either a lab environment or shipboard as necessary.
- m. The Contractor shall develop and design firmware for components or system blocks that meet specified requirements for radar distribution applications.
- n. The Contractor shall be responsible for performing and reporting on functional testing of software, including unit testing, integration testing, system testing, and acceptance testing.
- o. The Contractor shall also be responsible for performing and reporting on non-functional software testing, including security testing, performance testing, usability testing, and compatibility testing.
- p. The Contractor shall provide cybersecurity services for systems developed under this contract that enforce, comply with, and support Department of the Navy (DoN), Department of Defense (DoD), cybersecurity directives, policies, and procedures.
- q. Cybersecurity services shall include technical, functional, operational, and managerial services necessary to ensure the security of software.
- r. The contractor shall be responsible for ensuring software complies with DoDI 8500.01 requirements throughout design, development, and deployment.
- s. If vulnerabilities are discovered, the contractor shall be required to make all necessary changes to software in order to mitigate said vulnerabilities.
- t. The Contractor shall develop Version Description Documents (VDD), Interface Design Documents (IDD), Software Development Plans (SDP) and documentation that supports any software development performed.
- u. The Contractor shall provide software troubleshooting and resolution for specified software applications.

- v. The Contractor shall review, provide recommendations, and implement ECPs following government approval.
- w. The Contractor shall provide Objective Quality Evidence (OQE) to demonstrate all non-COTS have been tested.
- x. The Contractor shall only use licensed software that is approved by the government.
- y. The Contractor shall provide OQE to demonstrate that all non-Commercial-Off-the-Shelf (non-COTS) hardware and software components have been fully tested against all system requirements.
- z. The Contractor shall ensure that all custom-developed software and firmware is tested and verified in strict accordance with the requirements of the DISA Application Security and Development STIG.
- aa. The Contractor shall perform, at a minimum, static and/or dynamic code analysis testing on all custom-developed source code as part of the formal verification process.
- bb. The Contractor shall maintain design release baseline (also referred to as developmental configuration and release record) and baseline for each software version.

C.3.2.4 Future Systems Engineering

- a. The Contractor shall, when authorized via a separately issued and funded contract modification, provide the necessary engineering services to design, develop, prototype, test, and produce new radar distribution systems, subsystems, or components. These services include the creation of TDPs, engineering documentation artifacts, Integrated Logistics Support (ILS) artifacts, prototype fabrication, and all associated test activities for technical solutions for emergent requirements. No work on a new system shall be performed, and no costs shall be incurred under this section, until a formal contract modification has been executed by the Contracting Officer.

C.3.3 TECHNICAL SERVICES

C.3.3.1 Installation Support

- a. The Contractor shall perform shipboard and shore-based site Hardware and/or Software Installations at the direction of the Government.
- b. The Contractor shall perform shipboard and shore-based site Hardware and/or Software Installation Testing using Government approved Test Procedures.
- c. The Contractor shall perform repairs or replacement of installed equipment as directed by the Government.
- d. The Contractor shall perform environmental qualification testing (EQT) to include Electromagnetic Interference/Electromagnetic Compatibility (EMI/EMC) testing, vibration testing, mechanical shock testing, ingress protection testing, salt spray testing, sand and dust testing, and temperature and humidity testing.
- e. The Contractor shall provide EQT Object Quality Evidence (OQE) report for each of the testing listed in d above.
- f. The contractor shall perform all EQT testing in a Government approved facility or 3rd party approved vendor.

- g. The Contractor shall perform platform installation testing, to include Stage 1-7 testing in accordance with DOD-STD-2106A.
- h. The Contractor shall perform form, fit, and functional testing of replacement components, assemblies and sub-assemblies in support of sustainment engineering solutions.
- i. The Contractor shall develop and maintain test plans for production, prototype, and sustainment radar distribution systems.
- j. T The contractor shall develop and maintain test plans for EQT and platform installation testing.
- k. The contractor shall develop, deliver, and maintain a comprehensive Test and Evaluation Master Plan (TEMP).
- l. The contractor shall develop, document, and deliver all required test plans, ensuring each includes the overall test strategy, detailed procedures with specific test case scenarios, a requirements traceability analysis, and a data management and analysis plan.
- m. The contractor shall record test data in a format suitable for analysis and inclusion in a test report.
- n. The contractor shall use test, simulation, and/or model data with the goal of discovering useful information, informing conclusions, and supporting decision-making recommendations about current radar distribution applications.
- o. The contractor shall monitor the results of the test and compare them with the expected results.
- p. The contractor shall prepare and deliver a completed test report at the conclusion of the test event to the Government. The test report shall include a summary of all test activities and final test results of an installation, development, or updating a system project.
- q. The contractor shall provide reclamation support to test, repair, certify, and package reclaimed assets from afloat platforms and shore sites.

C.3.3.2 Sustainment Support

- a. The Contractor shall provide technical assistance at distance and/or on-site for casualties to fielded systems as directed by the Government.
- b. The Contractor shall perform repairs or replacement of existing equipment as directed by the Government.
- c. The Contractor shall build, fabricate, assemble or procure components and systems in accordance with the specifications and requirements of a Government approved Technical Data Package (TDP).
- d. The Contractor shall perform lifetime buys of End-of-Life (EOL) components as part of the system's obsolescence management. All procured components must be specified in the Government-approved Technical Data Package (TDP), and each procurement requires prior authorization from the Government. The Contractor shall coordinate procurement actions with Naval Supply Systems Command (NAVSUP) and/or Defense Logistics Agency (DLA).
- e. The contractor shall repair or assemble PCBs and CCAs. This shall include inspection, troubleshooting, desoldering, soldering, and testing.

- f. The Contractor shall test, assemble and integrate radar distribution systems with other existing systems.
- g. The Contractor shall assemble cables, impedance-controlled cables, harnesses, and wiring harnesses for a Government approved TDP for radar distribution systems.
- h. The Contractor shall mechanically assemble units, enclosures, or systems by reading, updating, or following mechanical drawings, industry methods, or assembly instructions.

C.3.4 CONFIGURATION MANAGEMENT AND INTEGRATED LOGISTICS SUPPORT

- a. The Contractor shall provide Integrated Logistic Support (ILS) support for the life cycle management of radar Distribution System equipment.
- b. The Contractor shall maintain revised and/or updated data products to reflect the latest Government approved baseline.
- c. The Contractor shall perform Configuration Management (CM) and provide ILS encompassing product support planning, supply support, provisioning, planned and Reliability Centered Maintenance (RCM), PHS&T, training, and the development of technical manuals using the S1000D process.
- d. The Contractor shall establish and maintain a CM system to provide control and traceability over all items defined in the TDP. This system shall manage all hardware, software, firmware, and documentation, ensuring the accurate tracking of all system configuration baselines.
- e. The Contractor shall develop and deliver a Configuration Management Plan (CMP) that defines, organizes, and establishes all processes, procedures, and support systems required to implement the following configuration management disciplines: CM Planning, Configuration Identification, Configuration Control, Interface Management, Configuration Audits, and Configuration Status Accounting.
- f. The contractor shall develop, update, and deliver a Life Cycle Sustainment Plan (LCSP) that provides technical input to ensure sustainment considerations are integrated into all planning, implementation, and management activities associated with the acquisition, development, production, fielding, support, and disposal of supported radar distribution systems across their life cycle.
- g. The contractor shall be responsible for the configuration control of the developmental configuration and may iteratively design, release, prototype, and test until the functional and allocated requirements are satisfied.
- h. The Contractor shall ensure the developmental configuration is defined by a complete set of released and government approved engineering documentation, to include all hardware drawings, software design documents, interface control documents, and test and verification documents.
- i. The contractor shall define product structure identifying Configuration Items (CI) and configuration documentation.
- j. The Contractor shall assign CI identifiers, part, item, software identifiers, and nomenclature and determine and assign type of specification(s) for each CI.
- k. The Contractor shall approve CI performance or detail specification for each CI for which the contractor has configuration approval authority, establishing a (contractor) allocated baseline.

- l. The Contractor shall define traceable items and prescribe method of tracking identification (serial or lot control).
- m. The Contractor shall provide input through the CCB process for design changes to the prototypes and systems, and properly adjudicate any associated ILS product impacts.
- n. The Contractor shall provide an Indentured BOM for prototype systems.
- o. The contractor shall develop and provide a Level of Repair Analysis (LORA) compliant with SAE AS 1390.
- p. The Contractor shall perform technical reviews and engineering assessments in support of developing and maintaining any radar distribution systems.
- q. The Contractor shall provide logistics data for affordability, reliability, maintainability, supportability, supply support, maintenance concepts, and warranty considerations.

C.3.4.1 Preventative Maintenance System Development and Updates

- a. The Contractor shall provide technical expertise to resolve PMS Feedback Reports for the radar distribution systems equipment.

C.3.4.2 Allowance Parts List (APL) Development and Updates

- a. The Contractor shall develop APL for future Distribution Systems equipment.
- b. The Contractor shall provide provisioning data to update APLs for any changes to the approved configuration baseline.
- c. The Contractor shall provide technical input for engineering support requests.

C.3.4.3 Training Development and Updates

- a. The Contractor shall establish and update data requirements to support the life cycle maintenance of all radar Distribution System equipment training data products in accordance with MIL-PRF-29612B.
- b. The Contractor shall establish or update data requirements to support the life cycle maintenance of existing training data products in accordance with MIL-PRF-29612B.

C.3.4.4 Interactive Electronic Technical Manual (IETM) Development and Updates

- a. The Contractor shall provide operation and maintenance technical manuals for all supported radar distribution systems and equipment using the S1000D format, legacy 2RI format, or a new future format.

C.3.5 OBSOLESCENCE MANAGEMENT

- a. The Contractor shall provide Obsolescence management to identify, report, and mitigate risks associated with Diminishing Manufacturing Sources and Material Shortages (DMSMS) for components.

- b. The Contractor shall ensure the obsolescence required changes for production, and/or repair of the equipment shall be made in accordance with the Government approved CM Plan.
- c. The Contractor shall develop and provide a DMSMS Plan in accordance with TA-STD-0016 and shall analyze, report, and execute obsolescence management for the prototype system and equipment; investigate, evaluate, develop, and replace, as applicable, of obsolete or non-obtainable parts and components for all supported systems to mitigate program supply chain risks.
- d. The contractor shall manage obsolescence to ensure compliance with all performance and contract requirements.
- e. The Contractor shall establish and maintain a formal process to identify pending and emergent obsolescence issues and vendor-implemented changes and shall immediately notify the Government of any such issues affecting equipment within the approved baseline.
- f. The contractor shall report instances of DMSMS.

C.3.6 CYBERSECURITY SUPPORT

- a. The Contractor shall develop Risk Management Framework (RMF) packages as the ISO/ISSE and Navy Qualified Validator (NQV) for the radar distribution product suite. RMF efforts shall include development of Assess Only Determination requests and the follow-on Assess Only Authorization packages. If a system is not a candidate for Assess-Only, the Contractor shall develop a full RMF package. Any RMF package development shall be consistent with published guidelines available from the NAVSEA Functional Authorizing Official (FAO).
- b. The Contractor shall conduct Annual Security Reviews (ASR) for each RMF package to include all testing, RMF package updates, maintenance of the system Plan of Actions and Milestones (POA&M) and addressing any conditions of the Authority to Operate (ATO).
- c. The Contractor shall coordinate assessment activities for all identified Cross Domain interconnections.
- d. The Contractor shall provide cyber assessment and continuous monitoring support for all product baselines to include performing assessments on a quarterly basis for all applicable DISA STIGs, Security Content Automation Protocol (SCAP), and conducting vulnerability assessments with various tools to include the Assured Compliance Assessment Solution (ACAS). The assets to be supported include Red Hat Enterprise Linux, network services and devices, and all hardware and software covered under this solicitation.
- e. The Contractor shall provide technical assessments for all discovered vulnerabilities and recommend corrective solutions to resolve them. Discovered vulnerabilities will require updating and maintaining the system Plan of Actions and Milestones (POA&M).
- f. The Contractor shall perform System Manager functions for all associated Video Random Access Memory (VRAM) system baselines on Secret Internet Protocol Router (SIPR) for the Sensor Distribution Systems (SDS) product suite baselines. System

Manager functions include answering all cyber directives, maintaining the site associations, asset assignments, uploading system baseline images on baselines, reviewing and updating remediation details for all vulnerabilities, and discovering and correcting any anomalies in the system records.

- g. The Contractor shall provide technical planning, design analysis, evaluation of alternatives, and systems engineering to ensure cyber requirements are met with respect to availability, integrity, authentication, confidentiality, and non-repudiation of critical system information.
- h. The Contractor shall provide analytical and technical security recommendations to other team members.
- i. The Contractor shall support registration of systems in DoD IT Portfolio Repository- Department of Navy (DITPR-DON) and registration of applications within DON Application and Database Management System (DADMS).
- j. The Contractor shall support the registration requirements and updates for the Ports, Protocols, and Services Management (PPSM).
- k. The Contractor shall develop and deliver all documentation required to support the RMF, consisting of the Privacy Impact Assessment, e-Authentication documentation, Standard Operating Procedures (SOPs), Continuous Monitoring Strategy, Security and Security Assessment Plans, Risk Assessment, and all system Network Diagrams. The Contractor shall support analysis of cyber requirements for the Combat System Element (CSE) with respect to meeting Combat System (CS) Enclave level requirements and integration efforts.
- l. The Contractor shall develop all related STIGs and security testing artifacts such as vulnerability scans, fuzz testing, POA&M, and System Security Authorization and Operations Management (SSAOM).
- m. The Contractor shall provide all details on ports, protocols, and services used for any current or future radar distribution system.
- n. The Contractor shall perform patching of the operating system and conduct system hardening in accordance with the latest DoD and DoN requirements.
- o. The Contractor shall maintain ship support personnel that meets Cyber Security Workforce (CSWF) requirements.
- p. The Contractor shall maintain ship support personnel that meets privileged access requirements.
- q. The Contractor shall ensure system design and implementation are compliant with the requirements of the DFARS for safeguarding covered defense information.